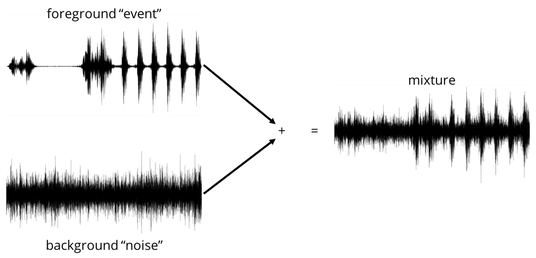
Segregation from Noise as Outlier Detection

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Introduction

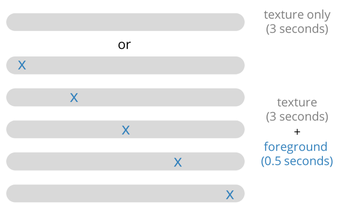


- The auditory system must segregate foreground events from noise in order to make sense of the everyday acoustic environment
- **Do listeners identify foreground sound events by estimating distributions over environmental background sounds and registering outliers of these distributions as new events?**

Experimental Task

- Listeners detected brief (0.5 s) foreground sounds embedded in real-world background noise (3 s excerpts of sound textures)

"Did you hear ONE SOUND (the continuous background sound only) or TWO SOUNDS (the continuous background sound AND some brief additional sound)?"



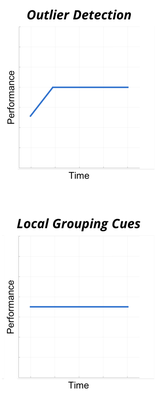
- All data was collected via Amazon Mechanical Turk

Experiment 1: Hypotheses

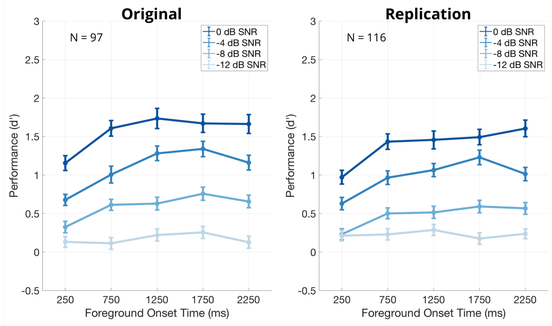
- Listeners detect acoustic events as outliers under a model of an environment's background texture
- Prediction: Ability to detect events will increase with longer exposure to background texture

vs

- Foreground sounds just segregated via local grouping cues
- Prediction: Exposure to the background will have no effect on ability to detect events in noise



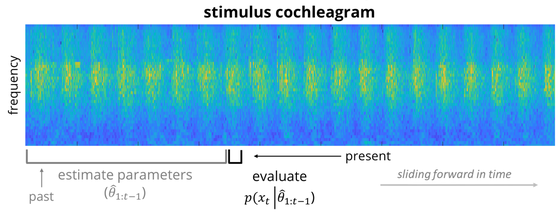
Experiment 1: Results



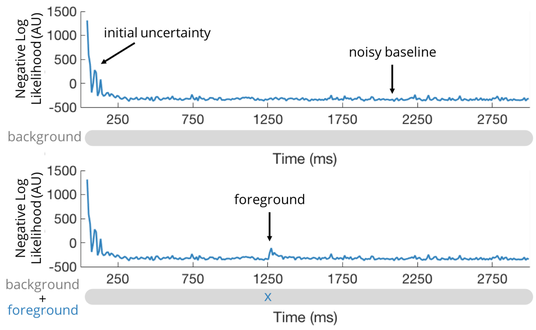
- Detection performance (d') increases with foreground onset time then plateaus
- Suggests listeners benefit from collecting more samples from background
- Consistent with idea that listeners estimate distribution of background noise and use to detect foreground events

Experiment 1: Modeling Methods

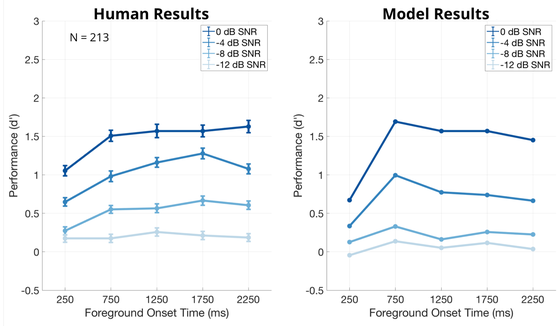
- Fit multivariate Gaussian to stimulus cochleagram in the past
- In a present window, evaluate likelihood of data given past fit
- Slide forward in time and repeat



- Outliers appear as spikes in negative log likelihood
- Model outputs exceeding threshold are classified as foreground sound events



Experiment 1: Modeling Results

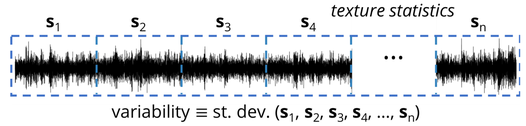


Experiment 2: Motivation

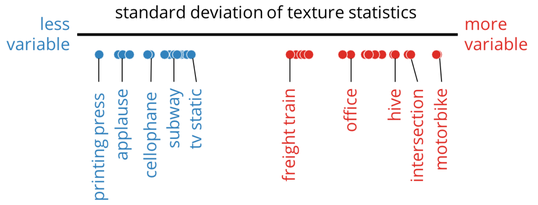
- Previous work suggests extent of temporal averaging window may scale with background variability
- Would more variable backgrounds require more data to accurately estimate the background distribution?

Experiment 2: Stimuli

- Measure texture statistics of backgrounds

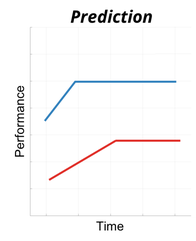


- Sort backgrounds by variability of statistics

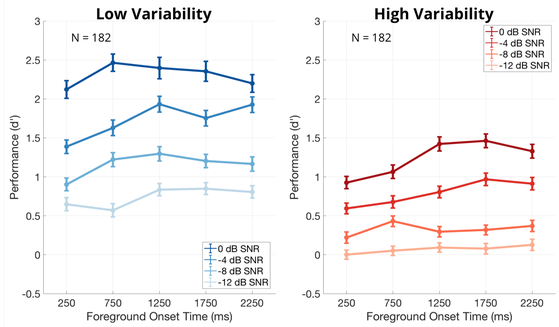


Experiment 2: Hypothesis

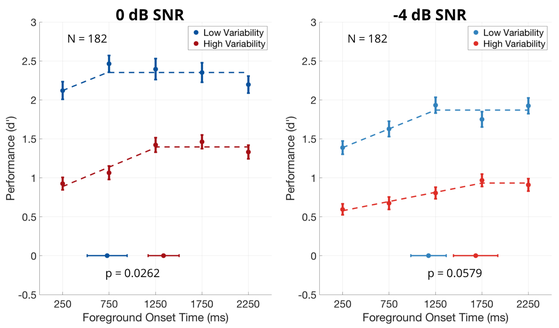
- More variable backgrounds require more data to accurately estimate background distribution
- Peak performance should occur later in time for more variable sounds



Experiment 2: Results



- Peak foreground detection performance occurred later when background sounds were less homogenous



- Suggests listeners collect a larger sample when more data is required to accurately estimate parameters of the background
- Work modeling behavioral effects is ongoing

Conclusions

- Foreground detection improved over the first one-second exposure to background
- Peak foreground detection performance occurred later when background sounds were less homogenous
- Results are consistent with the idea that listeners estimate the distribution of ongoing background noise and segregate sound events that are outliers from this distribution

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